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Mining educational data to predict students performance

A comparative study of data mining techniques

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Abstract
Information is everywhere in a hidden and scattered way. It becomes useful when we apply Data mining to extracts the hidden, meaningful, and potentially useful patterns from these vast data resources. Educational data mining ensures a quality education by analyzing educational data based on various aspects. In this paper, we have analyzed the academic results and behavior of some engineering students. For this study, we collect data from 80 students from the CSE department. We gather data from mark sheets and other relevant factors that accelerate the results, collected through a survey. Our main goal is to predict the students' performance. According to this prediction, the counseling department will guide them in advance so that those who are likely to have bad results can do better. The classification can be based on various aspects, as many factors improve the educational system. We have created two datasets focusing on two different angles. Our first dataset classifies and predicts the category of a student (good, bad, medium) on a specific course based on their prerequisite course performance. We have implemented this in the artificial intelligence course. Our second dataset also classifies and predicts the final grade (A, B, C) of any random subject, here we organize our data such a way where it will only focus on how their performance was till the midterm exam. We analyze and compare six classification algorithms. We have focused on all aspects of an algorithm, not only the accuracy level but also the complexity and cost. We have built two final models for two of our datasets based on a decision tree and the naive Bayes algorithms accordingly.

Keywords Data mining techniques · Ensemble learning · Model building · Prediction

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1 Introduction

Nowadays the necessity of extracting data of educational field is increasing day by day. Modern educational organizations help the decision makers to obtain the correct knowledge and to make best decisions by using new modern techniques such as data mining methods which is very important to develop and enhance educational system (Veeramuthu and Periasamy 2014). Educational data mining (EDM) is a branch of data mining and machine learning research to develop new ways to analysis educational data from an educational system (Pathan and et al. 2014). It is the process of converting raw data from educational institution to usable patterns (Tan et al. 2016). In this paper, we introduce how educational data mining has an impact on counseling students such a way where they can not only develop their academic result but also they can improve their post-graduate career.

We have collected data from two resources (mark sheet and survey). We have collected student's academic mark sheets from where we can analyze their performance course-wise and surveyed the same students. Besides the academic results, we have analyzed their educational behavior like students learning strategy. How they like to study their courses (slide, book, browsing the internet or combining them, etc.). We have also focused on the percentage of students wanted to build their career as a software engineer or as a researcher after graduation. The students should be guided according to their aim. The interest level on various courses and their practical learning activity besides academic study like (web development, graphics designing, android, programming practice, etc.). These factors have an impressive impact on developing a computer engineering student's career. If authority analyzes these factors besides the academic result both students and institute get facilitate. In this paper we have pre-processed our dataset and used WEKA machine learning tool for mining data, building model and interpreted them. We have applied six classification algorithms which include three basic algorithm decision trees, naive Bayes and PART and three ensemble algorithm bagging, boosting, and random forest. We have analyzed various resulting factors (i.e. confusion matrix, error rate, rms., etc.) of these classifiers and compare the accuracy level. We have seen that ensemble techniques always give high accuracy. But as we consider the complexity and the cost of an algorithm besides the accuracy level, in our system we preferred the decision tree model according to the first dataset and naïve Bayes model for the second dataset.

We have two types of datasets. We named our first dataset as AIprerequisit dataset as it is created based on the performance of prerequisite courses of artificial intelligence and we named our second datasets as AllTheoryPerformace. This dataset consists in a way where we predict students a final grade of any course based on their marks obtained till the midterm examination. The rest of the paper is organized in the following way. Section 2 represents some related works. Section 3 represents the methodology. Section 4 shows a comparative view of various classifiers and the experimental results. Finally, Section 5 concludes our work including future scopes of this research.

2 Related work

In 2011, R. R. Kabra and R. S. Bichkar predicted the engineering student's performance using a decision tree approach. They have identified the students who are likely to fail in advance by applying the decision tree algorithm. In this research data of 346 students is collected. They collected subject wise marks of SSC and HSC exam and also some personal information like gender, financial condition, etc. for their dataset. Finally, using the WEKA machine learning tool, a decision tree model is constructed which generate several understandable rules and predict which students will fail or pass. 69.94% of accuracy was achieved from this model (Kabra and Bichkar 2011). In this paper we have analyzed six classification algorithm and collect course wise results as well as some other factors strongly related to study. We make our final model based on decision tree for our dataset-1 and naive Bayes for dataset-2.

In 2011, Brijesh Kumar Bhardwaj and Saurabh Pal predict student performance using the naïve Bayes classification algorithm. The authors Identified different factors, which affect a student's learning behavior and performance during their academic career. They gathered data from different degree colleges. They used matlab. In this research, they find out the relationship between student related variables (gender, student category, family size, live in hostel or not, family and financial status, previous grades, etc.) and predict how these variables affect academic performance. Finally, they show that only self-effort is not enough to improve academic performance, other factors have got a significant influence on student's performance (Bhardwaj and Pal 2012). In our paper we have collected CSE students data from Notre Dame University Bangladesh and use WEKA machine learning tool. We classify and predict students final result.

In 2014, Asraful Alam Pathan et al. represent a decision tree biased mining model for developing students C programming skills. The author constructed 2 types of dataset one are Based on students' behavior and results and the other one is Based on questionnaires consist of various c programming related questions. They generated two decision trees DT1 and DT2 based on these two types of data sets. They Used the WEKA machine learning tool for their experiments and Found 87% accuracy (Pathan and et al. 2014). In this paper we focused on all CSE courses include data structure, algorithm, database, etc. in dataset-2.

In 2014, Raisul Islam Rashu1 et al. Used data mining approaches to predict students' outcomes, i.e., final grade in a particular course by overcoming the problem of the imbalanced dataset. The authors implemented several re-sampling techniques to balance the dataset and Show accuracy increase after resampling data for every classification techniques. Classification Techniques based on C4.5 Algorithm, Naive Bayes, Backpropagation Algorithm are used in this research and the accuracy level was (45.56%), (61.11%.), (63.89%) for these classifiers respectively. They used 3 Resampling Techniques: SMOTE (synthetic minority oversampling Technique), Random under Sampling (RUS) and Random over Sampling (ROS). WEKA & Rapid miner was the main tools in this research (Rashu et al. 2014). In this paper we used data preprocessing methods for prepare our datasets.

In 2017, Md. Hasibur Rahman and Md. Rabiul Islam conducted a study on Evaluate student's academic performance and the impact of different features of students on their academic performance. The classification models were constructed by using four different classification techniques. In this paper, Naïve Bayes, KNN, Decision Tree, and ANN, bagging, boosting, random forest algorithms are used to evaluate student's academic performance. Finally, the ensemble filtering method was used to evaluate the student's academic performance. The main contribution of this research was showing that ensemble algorithms will always give the highest accuracy (Rahman and Islam 2017). In this paper we analyze both base and ensemble learners and choose base learner classifiers for making model considering complexity of ensemble learner.

3 Methodology

Data mining is the process of transforming raw data into useful information. There are four basic steps in data mining. The steps are: data gathering, data preprocessing, data mining or applying classification algorithms, interpretation (Tan et al. 2016). Figure 1 shows the total methodology of our work.

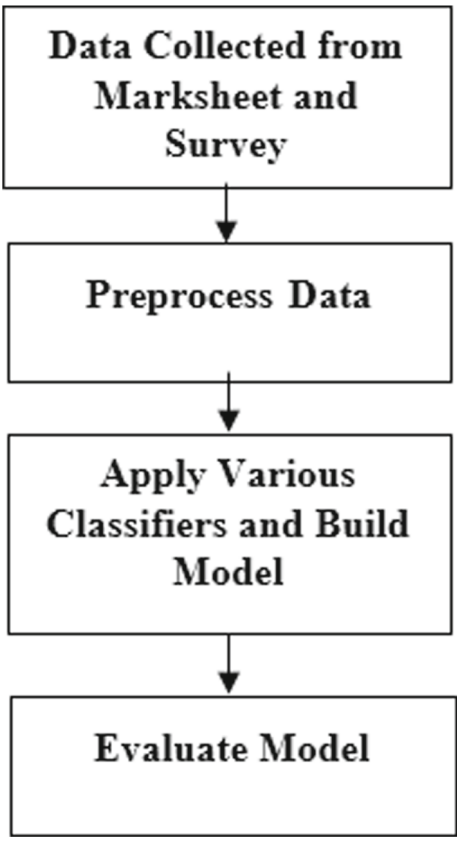


Fig. 1 Flowchart of Methodology

3.1 Data gathering

Data gathering can be accomplished by using a single resource or multiple resources. We have gathered 80 CSE student data. As Real Data always give the best outcome, we have collected real data of these students form two types of resources. We have collected students' mark sheets for obtaining the academic results of students also did a survey based on some behavior related to the performance. We have supplied a questionnaire containing some factors which help improve the educational system and counseling. Figures 2 and 3 shows the vital features of our questionaries and the features getting form the mark sheets.

3.2 Data preprocessing

The next basic step after data gathering is data preprocessing. As we know we have made our dataset from some real and raw data resources it is very normal that we have some incomplete, inconsistent and noisy data in our dataset like all other real-world datasets. Data pre-processing includes techniques that transform raw data into an understandable format and also make more suitable for the classifiers. Our classifiers' performance is dependent on how high-quality data we have and we can improve our data quality by preprocessing it.

There are several methods of data preprocessing like data cleaning, data integration, data transformation, data discretization, outlier removing, etc. below we will discuss some preprocessing methods we have used on our dataset.

– **Data Cleaning:** Data cleaning includes filling the missing values, smoothing the noisy data and resolving inconsistent data. We totally remove some of the missing values and some of them are replaced with an averaged approximated value (Alasadi and Bhaya 2017).

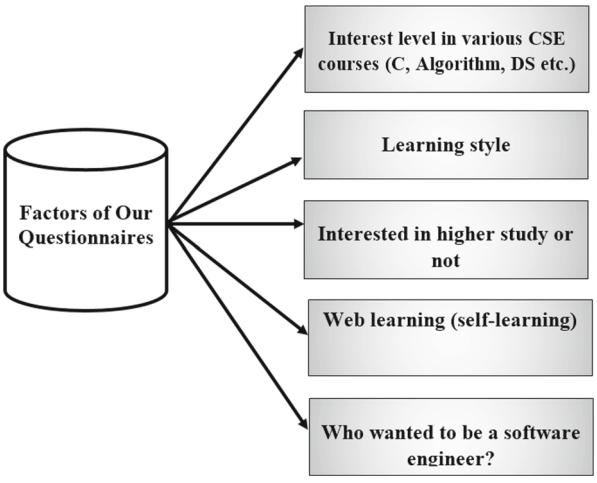


Fig. 2 Factors of survey

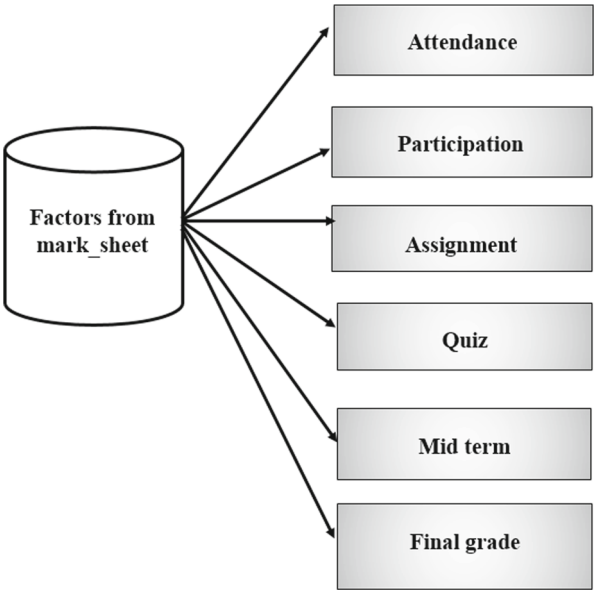


Fig. 3 Factors from marksheet

- **Data Integration:** Another data preprocessing method which includes merge data from various sources and represents a combined view to the user. We have combined data from our two data resources mark sheets and questionnaires in one dataset (García and Luengo 2015).
- **Data Transformation:** Data transformation includes normalization, aggregation, and generalization. We have transformed our fractional values into integer number by using "weka numeric transform" and convert some numeric value into nominal and categorical value i.e. (good, bad, medium, A, B, C, etc.) (García and Luengo 2015).
- **Data Discretization:** Discretization means discretize a range of numeric attributes in the dataset into nominal attributes. In this process, the total values are merged into small intervals and placed into bins so that the total number of states becomes low. The bins are considered as ordered and discrete values. Some specific algorithms like naïve Bayes Work efficiently after discretizing the numeric data. We apply discretization on our data with the help of weka discretization (García and Luengo 2015).
- **Outlier Removing:** Some time dataset contains some extreme values which do not match with the overall pattern of a sample. The classifier can not put them in any category or cluster. These types of data are called outliers. The accuracy increase after removing the outliers or fix those outlier values in the dataset. It may occur because of human, natural or sampling error. We detect and remove our outliers from the dataset by using wekaInterQuartileRange (García and Luengo 2015).

Table 1 Overview of datasets

Dataset	No.of attributes	Attribute types	Instances (train)	Instances (test)	Classes
Prerequisit- DataSet	13	Real & Nominal	40	28	3
FinalResult- Prediction	11	Real	240	130	5

– **Feature Creation & Selection:** In some cases, we have the information but not in a systematic way, in that case, we have to create features manually and put that information in a user understandable way. This process is known as feature creation.We have created some features (we learning level, interest level, learning strategy, etc.) from the information obtained from the questionnaire.Feature selection is a process by which we can understand which features are most important for classifying and predicting.We have done feature selection with the help of weka, using the ranking method (García and Luengo 2015).

3.3 Observe datasets and apply the right classifier

Every classifiers have its own advantages and drawbacks. No single algorithm is perfect 100%. In this paper we have applied six popular classification algorithm decision tree, naive bayes, PART, bagging, boosting and random forest. Each algorithm have its own speciality some works well on catagorical data types and some works well on continuous data. So, we need to choose classifier considering what kind of data

Table 2 Attributes description of dataset-1

Attribute name	Attribute values	Data types
C programming	Numerical	Continuous
C Programming Lab	Numerical	Continuous
Data structure	Numerical	Continuous
Data structure Lab	Numerical	Continuous
Algorithm	Numerical	Continuous
Algorithm Lab	Numerical	Continuous
OOP	Numerical	Continuous
OOP Lab	Numerical	Continuous
Interest level	High, Medium, Low	Categorical
Learning strategy	S,B,I,L	Categorical
Wanted to be a software engineer	Yes,No	Categorical
Web Learning Level	Good,Bad	Categorical
AI Theory	Good,Medium,Bad,Fail	Categorical
AI Lab(Target Attribute)	Good,Medium,Bad,Fail	Categorical

[Here we differentiate the grading system into 4 classes as: Good=(A+,A);Medium=(A-,B+);Bad=(B,C+,C,D);Fail=F grade for the attribute 'AI Theory' and 'AI Lab'(target)]

Table 3 Attribute description of dataset-2

Attribute name	Attribute values	Data types
Course	C Programming, Data Structure, Database, Algorithm, OOP, Networking	Categorical
Gender	Male,Female	Categorical
Attendance	Numerical	Continuous
Interest level	High,Medium,Low	Categorical
Learning strategy	S,B,I,L	Categorical
Interested in higher studies	Yes,No	Categorical
Web learning level	Good,Bad	Categorical
Participation	Numeric	Continuous
Assignmen	Numeric	Continuous
Quiz	Numeric	Continuous
Mid	Numeric	Continuous
Final Grade(Target Attr.)	A, B, C, D, F	Categorical

[In this table S,B,I,L represents Slide, Book, Internet and Lecture respectively and in target attribute we merge the grading system into 5 classes as: A=(A,A+,A-) ; B=(B,B+) ; C=(C,C+); D=(D) and F=(F)]

we have. Table 1 shows the overview of our two types of datasets and Tables 2 and 3 show the details of our datasets.

4 Result analysis

In this paper, we have applied six classifiers on our datasets and compare resulting factors. We have evaluated their accuracy level of prediction by applying them on our test datasets. We chose a classifier for building a final model for further evaluation. For building a final model we have consider both accuracy and complexity factors.

4.1 Analyze, evaluate and build model based on dataset-1

Table 4 is consrtucted based on training data of our dataset-1 from where we can see that random forest having 100% (best accuracy) and DT(J48), Naıve Bayes, PART having exactly same accuracy (87.5%) and ensemble algorithms bagging, boosting have (85%) accuracy. We can not accept 100% accuracy on train data set because of over fitting issue.

Now if we observe how all these six algorithm perform on test set (evaluation part), then we can see from Table 5 that, bagging algorithm giving the best accuracy (67.9%) which is an ensemble algorithm.Decision Tree, PART,Boosting algorithm giving the second highest accuracy (64.71%). Since ensemble algorithms(bagging,boosting) have some complexity issues, we tried to build a model by using a base learner(decision tree, naive bayes, PART).

Table 4 Apply various classifiers : A comparative analysis (Based on TrainData of Dataset:1)

Resulting factors	Decision Tree (J48)	Naive bayes	PART	Bagging	Boosting	Random forest
Accuracy (%)	87.5%	87.5%	87.5%	85%	85%	100%
Correctly classified instances	35	35	35	34	34	40
Incorrectly classified instances	5	5	5	6	6	0
TP rate	0.875	0.875	0.875	0.850	0.850	1.000
FP rate	0.115	0.084	0.123	0.180	0.180	0.000
Precision		0.860				1.000
Recall	0.875	0.875	0.875	0.850	0.850	1.000
F-measure		0.867				1.000
RMS	0.2277	0.2195	0.2357	0.2353	0.2731	0.1108
MAE	0.1037	0.0643	0.1111	0.1108	0.1956	0.0622
Kappa Statics	0.7647	0.7783	0.7602	0.7044	0.7044	1
ROC area	0.897	0.961	0.889	0.954	0.858	1.000

Table 5 Evaluate classifiers (Based on Train Data vs Test Data of Dataset-1)

Classifier	Accuracy (Train Data)	Accuracy (Test Data)
Decision tree	87.5%	64.2857%
Naïve bayes	87.5%	60.7143%
Part	87.5%	64.2857%
Bagging	85%	67.8571%
Boosting	85%	64.2857%
Random forest	100%	60.7143%

In this case, we choose Decision Tree for building a final model since it is the simplest algorithm among PART and Boosting, also generates some simple and understandable effective rules.On the other side if one want to use an ensemble algorithm ignoring all types of complexity, bagging algorithm will be best as it gives (67.9%) accuracy. Ensemble algorithms always perform well and give good accuracy.

4.2 Analyze, evaluate and build model based on dataset-2

In this section we have observed the same aspects as previous subsection A, but here we have applied the same classifiers on dataset-2. Table 6 has constructed base on train data.From where we can see that, random forest gives 100% accuracy and we ignore this for overfitting problem. Except random forest,we can see that PART algorithm giving highest accuracy (90.41%) and naive Bayes giving (80.83%) second highest accuracy on training dataset. We can also observe the other resulting factor from the following table.

Now if we apply the same classifiers on the test data of dataset-2, the scenario will change. Table 7 shows that boosting algorithm giving the highest accuracy (75.4%), bagging and naive bayes giving 74.61% and 73.1% respectively which differ by only 1%. In this case we choose Naive Bayes as a model for further evaluation considering the simplicity of its calculation and less complexity rather than the ensemble algorithms (bagging,boosting). If one wants to focus only the accuracy, and performance of classifier, Boosting model can be used.

4.3 Graphical analysis

Besides observing accuracy of models, we can find information by observing Threshold curve, ROC curve, classifiers errors. In this section we will observe ROC curve.

WEKA Explorer enables us to plot the ROC (Receiver operating characteristic) curve for a certain class label of dataset. This is important to evaluate the performance of our classifier.

Table 6 Apply various classifiers : A comparative analysis (Based on TrainData of Dataset-2)

Resulting factors	Decision Tree(J48)	Naive bayes	PART	Bagging	Boosting	Random forest
Accuracy (%)	86.67%	80.83%	90.45%	79.58%	66.67%	100%
Correctly classified instances	208	194	217	191	160	240
Incorrectly classified instances	32	46	23	49	80	0
TP rate	0.867	0.808	0.904	0.796	0.667	1.000
FP rate	0.080	0.097	0.060	0.126	0.220	0.000
Precision	0.871	0.809	0.907	0.793		1.000
Recall	0.867	0.808	0.904	0.796	0.667	1.000
F-measure	0.867	0.808	0.904	0.771		1.000
RMS	0.2059	0.2358	0.175	0.2424	0.3573	0.1025
MAE	0.0848	0.1012	0.0618	0.1421	0.2813	0.0599
Kappa Statics	0.7861	0.6906	0.8451	0.6551	0.4223	1
ROC area	0.947	0.941	0.976	0.957	0.764	1.000

Table 7 Evaluate classifiers(Based on Train Data vs Test Data of Dataset-2)

Classifier	Accuracy(Train data)	Accuracy(Test Data)
Decision tree	86.6667%	72.3077%
Naïve bayes	80.8333%	73.0769%
Part	90.4167%	70%
Bagging	79.5833%	74.6154%
Boosting	66.6667%	75.3846%
Random forest	100%	75.3846%

Figures 4 and 5 show two ROC curves, are based on decision tree (for dataset-1) and Naive Bayes (for dataset-2) for class value=Good and class value=A respectively. Here we having false positive rate (FPR) on the X-axis and true positive rate (TPR) has drawn on the Y-axis. In ROC curve the main goal is to have this curve lot more to the upper left corner which is "1". That means the closer the curve with y-axis, the better the classifier performs. Another evaluation through ROC curve is determined by AUC (Area Under ROC) which is clearly given in the graph. And both have pretty good percentage of AUC (Area Under Curve).

For decision tree ROC curve the area under roc is 97% and For naive bayes classifiers have 95% area under roc.

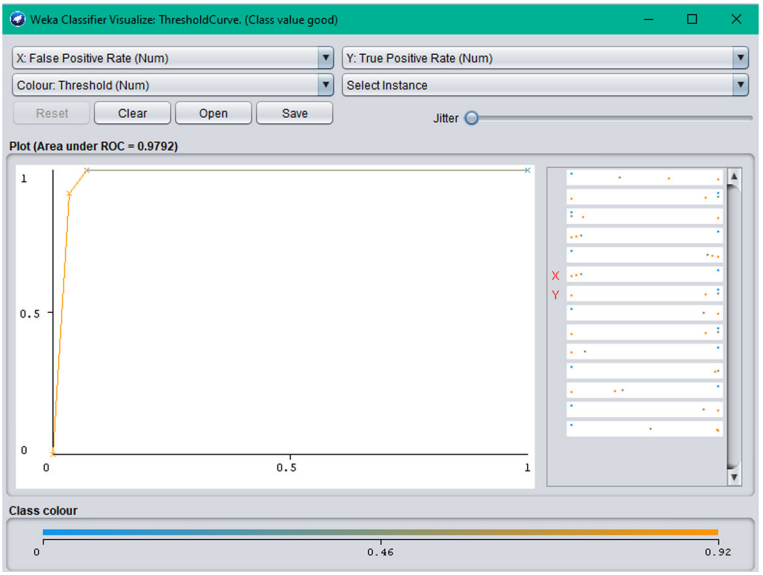


Fig. 4 ROC curve (Based on Decision Tree, Dataset-1, class value=good)

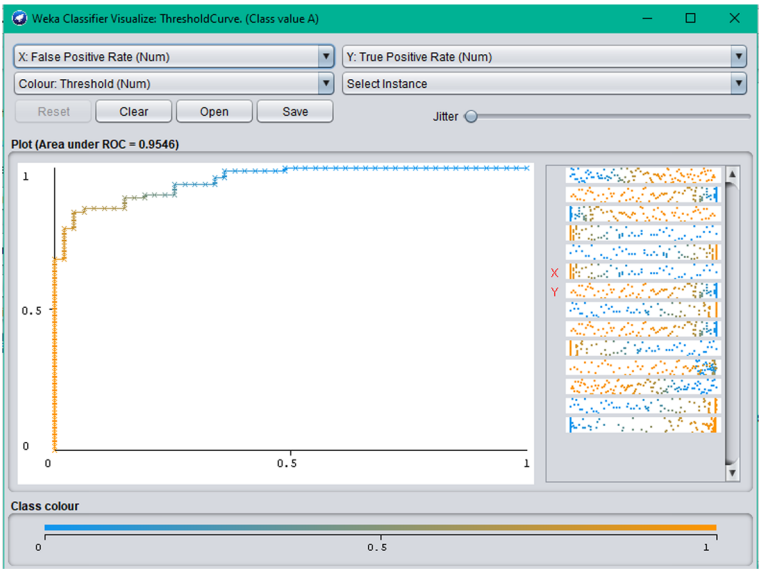


Fig. 5 ROC Curve (Based on Naive Bayes, Dataset-2, class value=A)

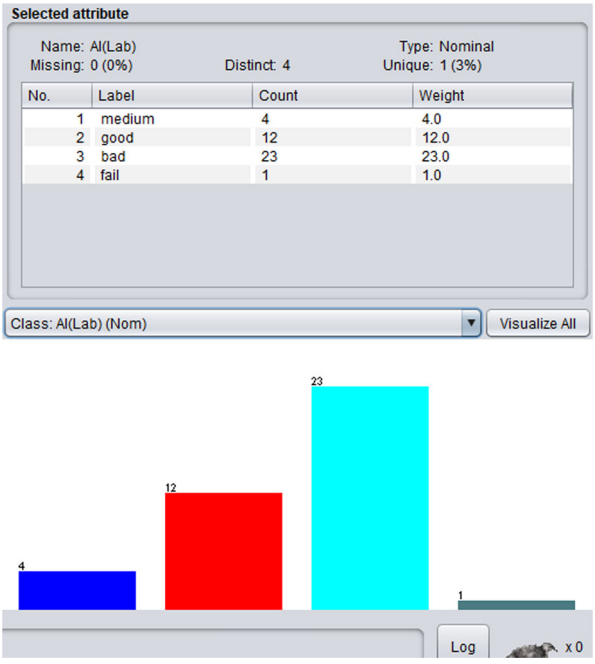


Fig. 6 Classifying train data (Data Set-1)

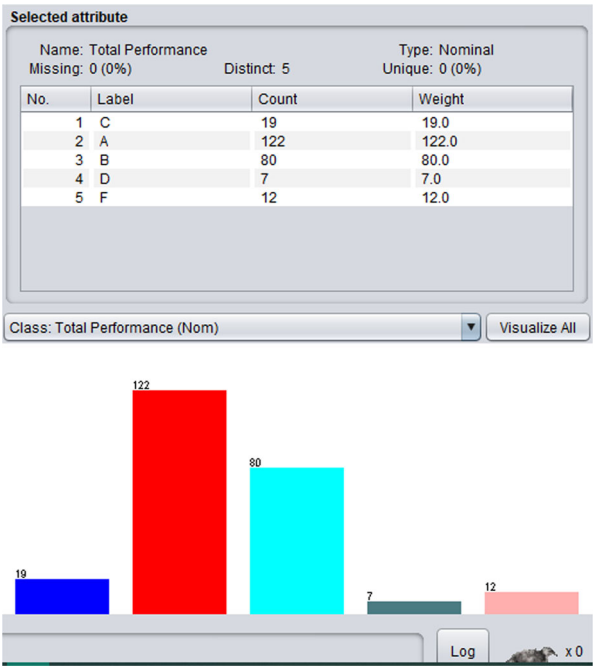


Fig. 7 Classifying train data (Data Set-2)

```
@relation prerequisit_test_Data_predicted

@attribute 'c programming' numeric
@attribute 'programming lab' numeric
@attribute 'data structure' numeric
@attribute 'data structure lab' numeric
@attribute algorithm numeric
@attribute 'algorithm lab' numeric
@attribute oop numeric
@attribute 'oop lab' numeric
@attribute 'web learning level' {good,bad,medium}
@attribute 'learning strategy' {SI,SBI,I,SBLI,SB,BLI,L,S}
@attribute 'wanted to be an engineer?' {no,yes}
@attribute 'final resul of AI(Theory)' {good,medium,bad,fail}
@attribute 'prediction margin' numeric
@attribute 'predicted AI(Lab)' {medium,bad,good}
@attribute AI(Lab) {medium,bad,good}

@data
75,72,60,66,82,75,65,71,medium,I,no,medium,-0.807692,good,medium
78,55,44,55,65,40,28,62,medium,SBLI,no,bad,0.807692,good,good
61,61,55,68,72,70,56,70,good,SBLI,yes,good,0.807692,good,good
63,50,12,0,66,45,36,61,bad,SB,no,fail,0.807692,good,good
50,53,60,65,80,52,50,63,bad,?,yes,good,0.807692,good,good
73,77,61,70,81,77,56,68,good,SBI,yes,medium,-0.884615,good,bad
65,73,68,76,88,76,70,78,good,?,yes,medium,0.807692,good,good
65,61,70,89,67,70,71,80,good,I,no,bad,0.807692,good,good
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Fig. 8 Prediction vs. Actual (Dataset-1)

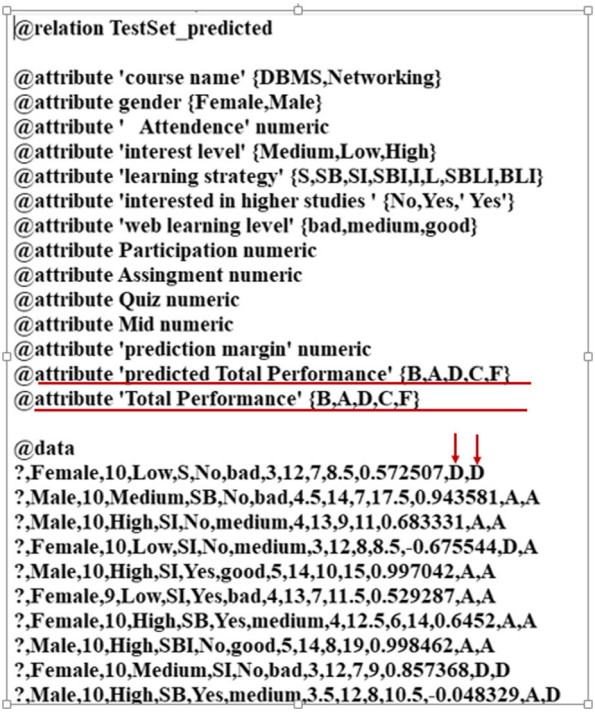


Fig. 9 Prediction vs. Actual (Dataset-2)

4.4 Visualizing classified data

In the following section we will see how data are classified for both prerequisite dataset and theory performance dataset. We can also see how classifiers predict the test dataset, comparing the actual value and the predicted value.

4.5 Prediction result

Figures 6 and 7 shows the classwise data based on prerequisit train dataset and finalResultPrediction dataset respectively. It shows the number of attributes under specific classes.

From Figs. 8 and 9 we can see the prediction result generated by classifiers and the actual result. Figure 8 based on prerequisit test dataset and Fig. 9 based on finalResultPrediction dataset. These figure shows that in most of the case actual class matches with the predicted result.

5 Conclusion

In this paper, information about academic result and student's behavior has been collected through their mark sheets and questionnaire from around 70 students. Two

kinds of datasets have been created to classify and predict student's results. The first dataset shows how student's results of a specific course can be hampered by the performance of it's prerequisite courses. The second dataset implies that the final result of any course is strongly connected with the performance till midterm exam. The sector-wise (attendance, quiz, assignment, participation, and midterm) marks till the midterm exam have been collected then we predict how will be their final grade in that course. This prediction will be very helpful for the student counseling division of any institution. From this, both students and the educational institution can get the best output. This study can also give a great impact on categorizing students according to their result, learning strategy and self-learning level from this counselor can change those student's point of view of learning. In this research, six classification algorithms have been analyzed and compared. Two models have been created from the analyzing result. The first one is based on the decision tree which is constructed by analyzing the first dataset named as AI Prerequisit dataset, gives 64.3% accuracy on the test dataset. The next model is based on the naïve Bayes created according to our 2nd dataset named Theory Performance and gives 75% accuracy on the test dataset.

In this paper, we have considered the data of only the CSE department of Notre Dame University Bangladesh. In the future, the same techniques can be applied for other departments as well as other institutions. Further study can be done by applying other data mining techniques like clustering and association rule to categorized the students and generate some efficient rules. We would like to collect more data to improve the accuracy of our classifiers and create a better model. In future the research can be made more realistic so that any educational institution can literally use it for their student's betterment.

Appendix

- **ANN:** ANN or Artificial Neural Network is a computational algorithm. This algorithm has been designed by using the concept of human neuron. It process information as like human brain analyze and processes information. It has the capability of self-learning that enables it to produce better results.
- **KNN:** KNN stands for K-Nearest Neighbour. It is a simple Machine Learning algorithm based on Supervised Learning technique. This algorithm find outs the similarity between new and existing data and puts the new data into the most category which is most similar with the available categories.
- **Weka:** An open source machine learning software with a collection of machine learning algorithms and data preprocessing tools. By using this software users can try out existing machine learning methods on their datasets in a flexible way.

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